

RESPIRATORY SYSTEM

★ It **consists of** the following parts:

A- Nose: It is formed of external nose and nasal cavity.

- **External nose:** Pyramidal in shape having a **root** and free **apex**, formed of **upper bony** part (**nasal bone**) and **lower cartilaginous** part.

- **Nasal cavity:**

- The nasal cavity is **divided** into 2 halves by a **nasal septum**.
- **Each half opens** on the face through the **anterior nasal opening** and opens **posteriorly in the nasopharynx** through the **posterior nasal opening**.
- Each nasal half of nasal cavity has **medial wall** which is the **nasal septum** and a **lateral wall** which has 3 bony projections called **nasal conchae** separating spaces called **nasal meatuses**.

- **Functions of the nose:**

1- **Smell.**

2- **Filtration and warming** of the inspired air.

3- Nasal conchae increase the surface area of the nasal cavity which magnifies the turbulence of air and thus **improves olfaction**.

- **Paranasal sinuses:** They are air-filled spaces in the skull bones surrounding the nose (pneumatic bones) and open in the nasal meatuses.

- **Functions:**

1- Decrease the weight of the skull.

2- Increase the resonance of voice.

3- They act as air cushions for the brain, eye and pituitary gland.

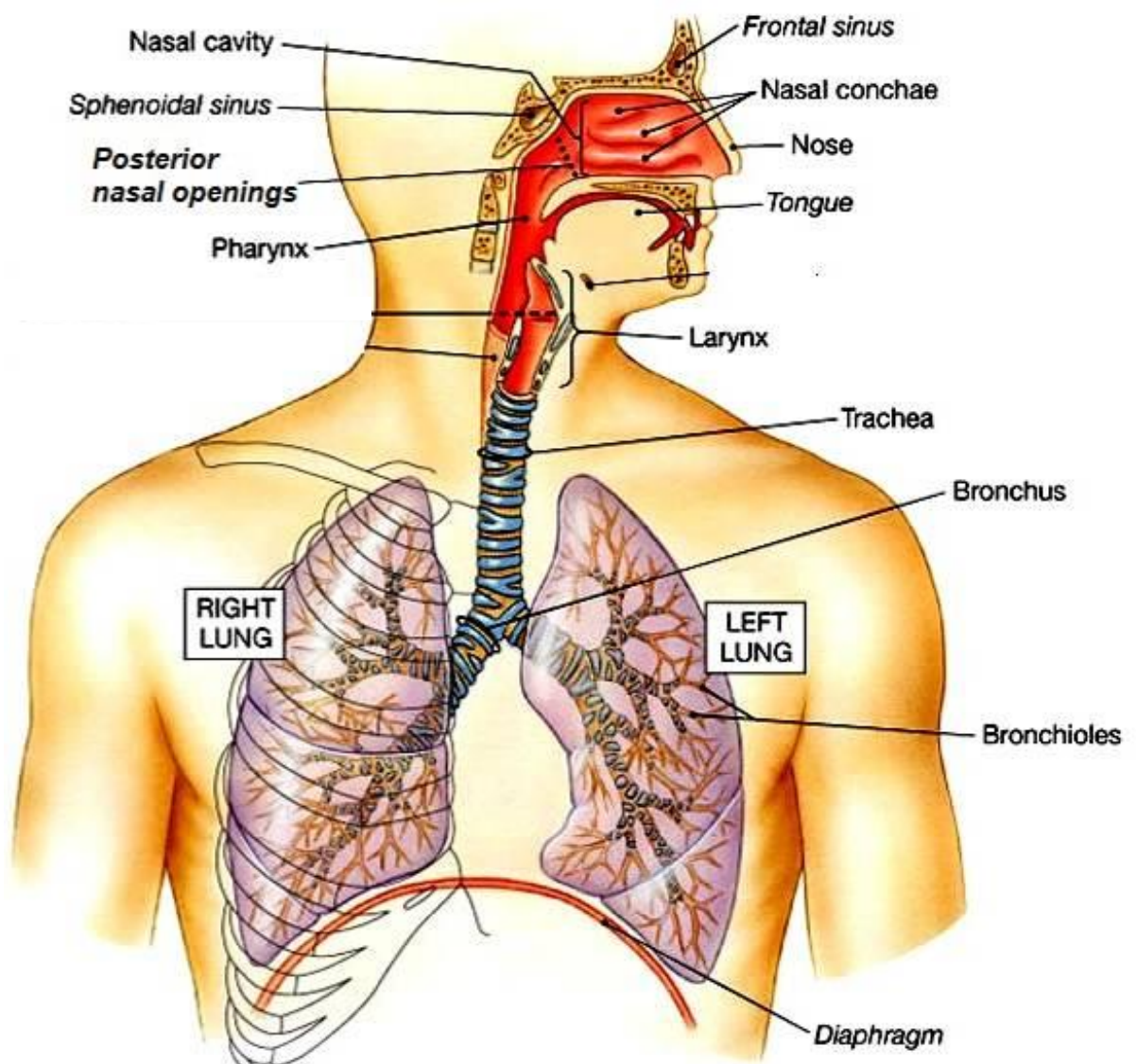
- **Paranasal sinuses are:**

1- Frontal sinus present in the frontal bone.

2- Maxillary sinus: The largest air sinus present in the maxilla.

3- Sphenoidal sinus: It occupies the body of the sphenoid bone below pituitary gland.

4- Ethmoidal sinuses: present in the ethmoid bone in the medial wall of the orbit.



B-Pharynx. (See Digestive System)**C-Larynx:**

- It is a tube consisting of 9 cartilages connected together by membranes and ligaments and moved by special muscles.
- It extends from the upper borders of the epiglottis till the beginning of trachea at the level of C6 vertebra.
- It contains the vocal cords which are responsible for production of voice.
- **Function:**
 1. Passage of air to trachea and lungs.
 2. Production of voice by vocal cords.
 3. Reflex expulsion of foreign bodies.

D-Trachea:

- 12 cm patent tube, transmits the air from the larynx to the lungs.
- It consists of 16-20 C-shaped cartilages.
- It lies in the midline, its upper 1/2 in the neck while its lower 1/2 in thorax.
- It ends in thorax at the level of T4 vertebra by dividing into two bronchi.

E- Lungs:

- Each lung is nearly half cone shaped structure having an apex above and base below and 2 surfaces (lateral convex and medial concave containing the hilum through which the structures enter and leave the lung).
- The **hilum** contains main bronchus, pulmonary artery and 2 pulmonary veins, autonomic nerves and lymph nodes.

- It is covered by the pleura which are closed sac formed of 2 layers: visceral and parietal with pleural cavity in between.
- **Difference between the 2 lungs :**

A- Right lung	B- Left lung
1. Short and wide 2. Formed of three lobes (upper, middle and lower) 3. Has two fissures (oblique-horizontal) 4. Has 10 segments 5. Has no cardiac notch	1. Long and narrow 2. Formed of two lobes (upper and lower) 3. Has one fissure (oblique) 4. Has 8-10 segments 5. Has cardiac notch

Lungs

